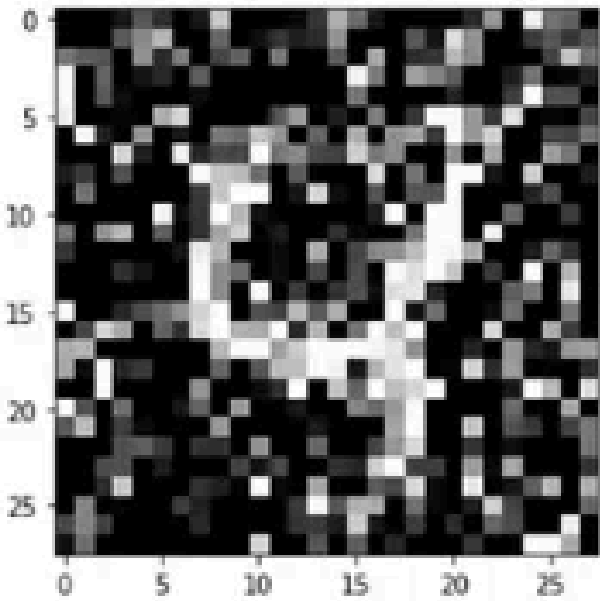
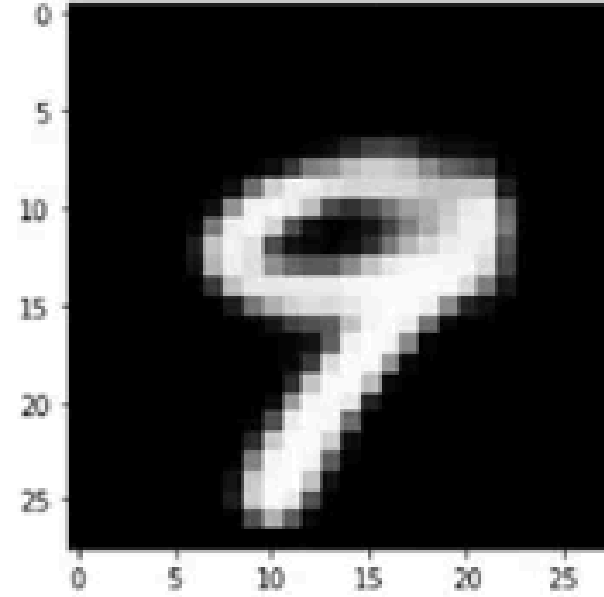
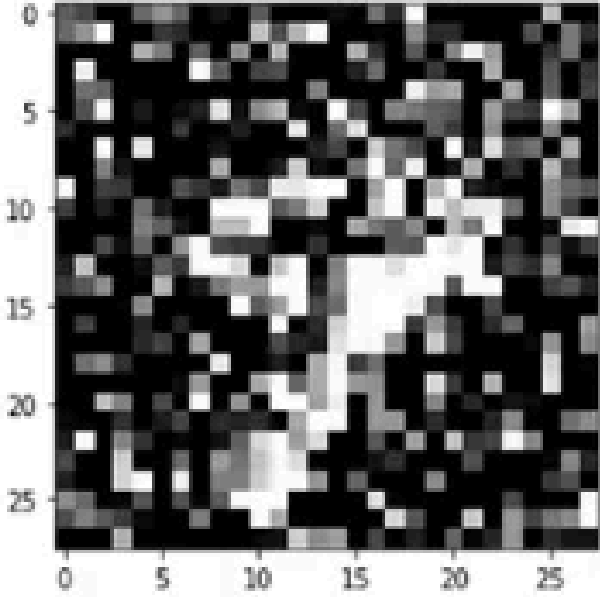
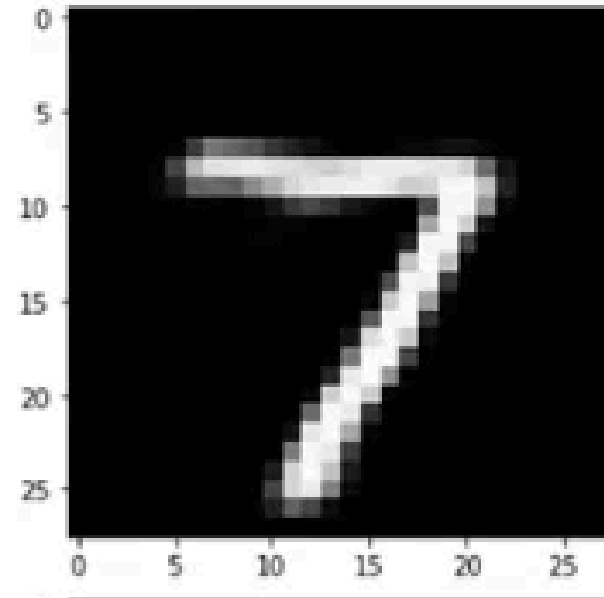
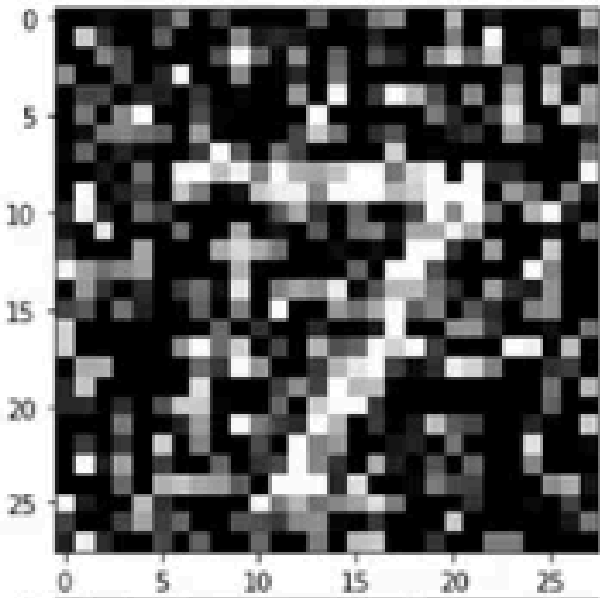
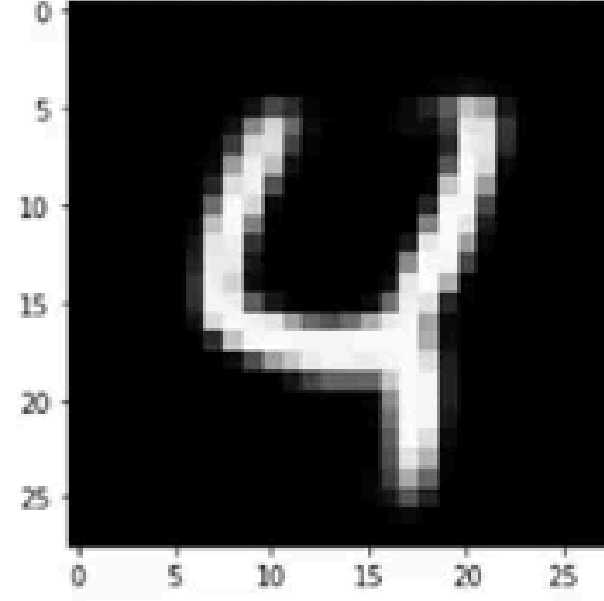


Corrupted image



Predicted image



Autoencoders in Multi-Omics Data Integration

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What is an Autoencoder?

An autoencoder is a neural network that learns a compact representation of input data and reconstructs the input from that representation.

01

Encoder

Maps high-dimensional input to a low-dimensional latent vector (bottleneck).

02

Latent Space

Compressed representation capturing essential structure; reduces noise.

03

Decoder

Reconstructs the original input from the latent representation; training minimizes reconstruction loss.



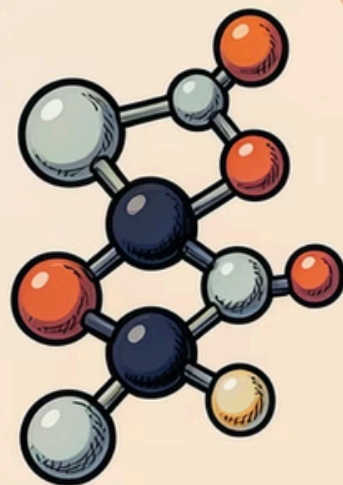
GENOMICS



Transcriptomics



PROTEOMICS



METABOLOMICS

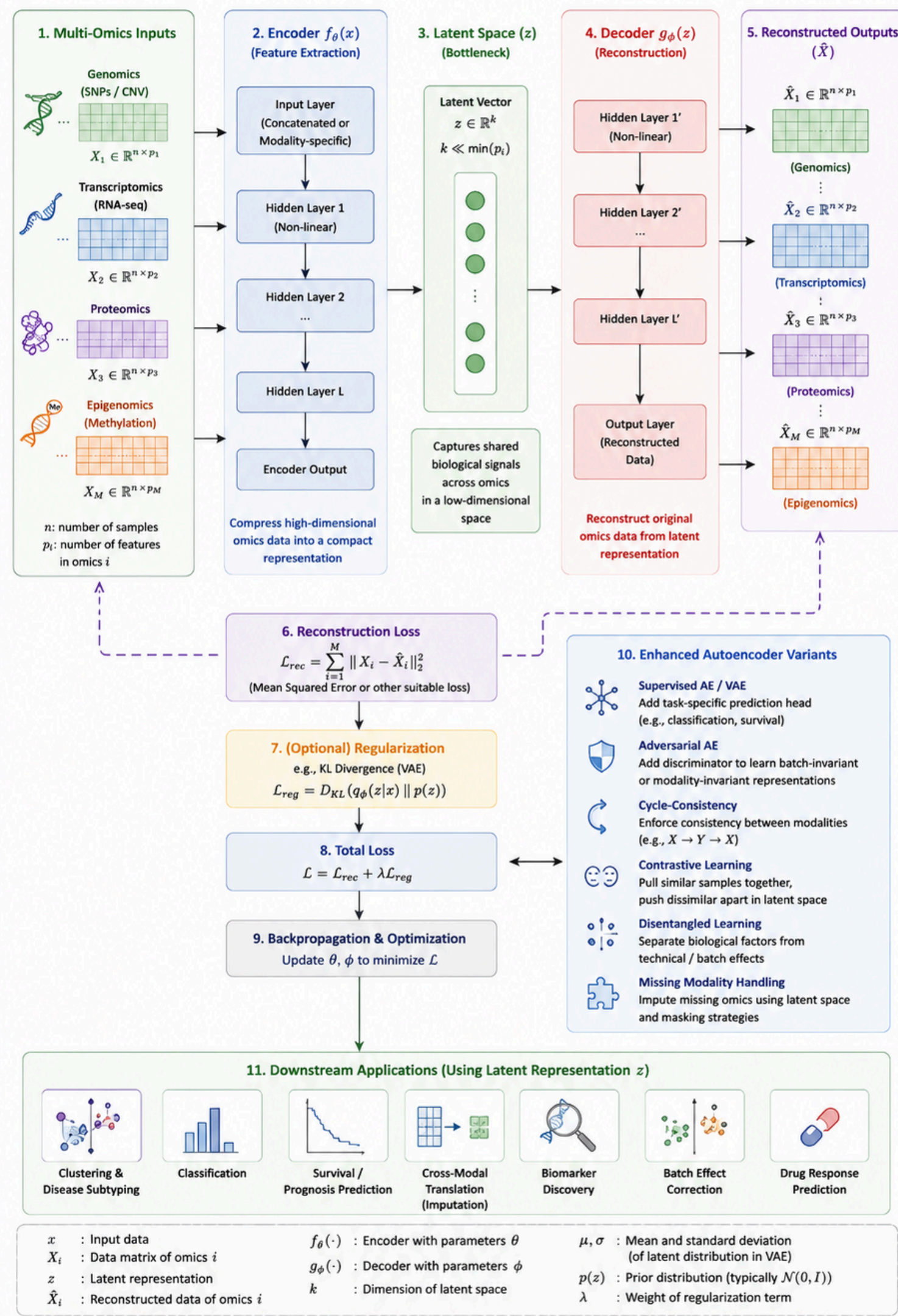
The Multi-Omics And Why Autoencoder

- Massive, heterogeneous datasets spanning DNA, RNA, proteins, metabolites.
- Frequent missing modalities and irregular sampling across patients.
- Complex, non-linear biological interactions that confound linear methods.
- Traditional stats (PCA, linear models) often miss key signals → need deep learning.

Autoencoders provide flexible, non-linear compression and reconstruction suited to multi-omics integration.

Autoencoders for Multi-Omics Integration

- Each omics modality is encoded (separate input branches or concatenated features) and merged into a shared latent space.
- Latent vectors capture cross-modal biological signals (pathways, cell states).
- Decoders reconstruct modality-specific outputs or jointly predict downstream labels.
- Advantage over PCA: learns non-linear relationships and hierarchical features.





Variational Autoencoders (VAEs)

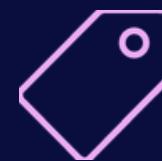
- VAEs learn a probability distribution over latent space (mean μ , std σ) instead of fixed vectors.
- Reparameterization trick: $z = \mu + \sigma \cdot \epsilon$ enables backpropagation through sampling.
- Loss = Reconstruction Loss + KL Divergence (regularizes latent distribution).
- Benefits: robust to noise, supports generation of synthetic samples, provides uncertainty estimates.

Key VAE Improvements (Overview)



MMD Regularization

Replaces KL with MMD for improved latent structure (e.g., MMD-VAE, MultiVI).



Supervised / Semi-supervised VAEs

Incorporate disease labels to improve classification and clinically relevant embeddings (OmiVAE, XOmiVAE).



Multimodal Integration

PoE / MoE fusion strategies to combine diverse omics (scMVAE, Multigrade).



Adversarial Learning

Discriminators remove batch effects and harmonize cohorts (GLUE, SCIM).



Contrastive Learning

Pulls similar samples together in latent space for robust clustering (MOSA).



Disentangled Learning

Separates biological signal from technical noise (scDisInFact, MIDAS).

PAIRLE TEN BINMPLEES:

CLINIIL DOAAITES

MPCNY	PATIENT	M9KIISK	(OBL)	S6PΔS	QUIES	MODALITIES
3010						
2000						
1280						
1280						
2020						
1940						

Handling Missing Data — Mosaic Integration

- Real cohorts often lack complete modality coverage per sample (mosaic missingness).
- Strategies: masking missing inputs during training, zero-padding, and prior-based imputation.
- Models like Multigrade, scVAEIT, MIDAS explicitly model missingness and impute modalities.
- Benefits: learn from partial records, improve robustness for clinical cohorts, enable modality prediction.

Popular Autoencoder Models – Comparison

Model	Main Idea	Use Case
OmiVAE	Supervised VAE	Cancer classification
MMD-VAE	MMD regularization	Cancer subtype prediction
OmiEmbed	Multi-task VAE	Survival prediction
scMVAE	Multimodal VAE	Clustering & imputation
Multigrade	Product-of-Experts + MMD	Mosaic integration
GLUE	Graph-based generative	Gene regulatory inference
MIDAS	Disentangled representation	Batch correction

Table highlights design choices and primary applications for commonly used models.



Key Challenges

- **High computational cost:** GPU needs, large memory footprint, lengthy training.
- **Limited labeled multi-omics data:** small cohorts increase overfitting risk.
- **Interpretability:** latent dimensions are abstract and require biological post-hoc analysis.
- **Hyperparameter tuning:** latent size, loss weights, and architecture choices critically affect results.

Future Directions & Conclusion

Foundation Models

Large pretrained biological models (scGPT, scFoundation) can transfer knowledge across datasets.

Beyond Omics

Integrate omics with imaging, EHRs, and wearable data for richer patient models.

Scalable Missing-Data Methods

Robust techniques for large clinical cohorts will improve translational impact.

Improved Interpretability

Methods to map latent features to pathways and clinical biomarkers will increase trust.

Autoencoders and VAEs are central to modern multi-omics integration — they enable disease prediction, biomarker discovery, and personalized medicine when combined with careful validation and interpretability methods.



Thank You

